Mohamed Jamal

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SUMMARY

Software Engineer with experience building scalable, high-performance applications. Skilled in Java Spring Boot and Angular, with a strong focus on optimizing systems, automating workflows, and improving overall efficiency. Passionate about problem-solving, writing clean code, and building reliable software. Always learning and refining my skills to stay ahead of industry best practices.

EXPERIENCE

Software Engineer **QNB Life Insurance** *Elmohandseen, Giza, Egypt* *05/2026 – 08/2026*

- **Legacy Code Maintenance:** Performed code tracing, bug fixes, and minor refactoring on monolithic Java applications (Servlets / JSP).
- **API & Logic Validation:** Assisted in validating backend data logic and updating service integrations to support internal system modules.

PROJECTS

Book Social Network *(Spring Boot, Angular, PostgreSQL ,JPA,Spring Security, JWT, Docker)*

- Built a full-stack social platform for book lending with an Angular front end and a Spring Boot REST API secured with JWT-based authentication and role management.
- Implemented the book lifecycle end to end: listing, shareable/archived status toggles, and a borrow → return → owner-approval workflow with transaction history tracking.
- Added a feedback and rating system for borrowed books, plus cover-image upload/storage through a dedicated file service and account email verification (Thymeleaf + Spring Mail, tested with MailDev).
- Working with pagination for filterable, paged catalog and history endpoints, and containerized the PostgreSQL/mail dependencies with Docker Compose.

E-Commerce *(Spring Boot, JPA, PostgreSQL, Docker, Flyway, Spring Security,CI/CD)*

- Designed and built a role-based e-commerce backend (USER/ADMIN) with JWT authentication and Spring Security, using MapStruct for clean entity-DTO mapping.
- Implemented order processing with real-time stock validation and automatic inventory deduction on purchase, including rollback logic on order deletion.
- Built a time-based order-locking rule that freezes orders 2 days after creation, preventing edits or cancellations past that window.
- Set up GitHub Actions CI/CD pipelines (separate build and deploy workflows) with Docker Compose for local and production environments.

Chat *(Spring Boot, WebSocket, Keycloak, OAuth2, JPA, PostgreSQL)*

- Built a real-time one-to-one messaging backend with instant delivery over **WebSockets** , including per-message sent/seen state tracking and support for media attachments (images, video, audio).
- Integrated **Keycloak** as an external identity provider, securing the API with OAuth2/JWT-based authentication.
- Designed a per-user **WebSocket notification** channel for live push updates on new messages and read receipts.
- Modeled the data layer with Spring Data **JPA** and Hibernate against PostgreSQL.

To-Do *(Angular, Spring boot)*

- Built a full-stack task management app with an Angular frontend consuming a Spring Boot REST API, supporting task creation, status updates (To Do / In Progress / Done), and deletion.
- Structured the frontend with a typed service layer and reactive HTTP calls (RxJS Observables) cleanly separated from UI components.

Twitter Clone (*Spring Boot, Spring Cloud, PostgreSQL, Kafka, Docker)* — *In Progress*

- Structuring the system as independent microservices: user service, centralized config server, and Eureka-based service discovery.
- Planning a Spring Cloud API Gateway for routing and Kafka for async communication between services as the next phase.

EDUCATION

Benha University — Faculty of Computers and Artificial Intelligence ***10/2019 – 06/2023***

- Major: Computer Science (CS)
- Grade: Good

Skills

- **Languages:** Java · JavaScript · C++ · Python
- **Backend:** Spring · Spring Boot · Hibernate · JPA · JDBC · RESTful APIs · WebSockets · RabbitMQ · JSF ·
- **Frontend:** Angular · HTML · CSS
- **Databases:** PostgreSQL · SQL · MongoDB · Oracle
- **DevOps & Tools:** Docker · Git · GitHub · CI/CD · Flyway · Linux · SVN
- **Security:** Keycloak · JWT · Spring Security
- **CS Fundamentals:** OOP · OOAD · Data Structures · Algorithms · Design Patterns · Problem Solving · Clean Code · Unit Testing · Web Development.

ADDITIONAL INFORMATION

Military Service: Completed

Languages: Arabic (Native), English (C1)